

PART B

Tarpack Gallery is a prestigious gallery that has been showcasing and selling artworks in Shah Alam since 2000. The gallery features a wide range of art pieces, including paintings, sculptures, photographs, and digital art. To modernize its operations, Tarpack Gallery has decided to transition from its manual record-keeping system to an online database system. They have chosen you to develop this new system. Below are the requirements and business rules for the database:

- The system must record customers' information, including their name, address, telephone number, and email.
- The system must track artworks' details, such as artwork ID, title, creation year, price and artist's identification.
- The system must store information about artists, including their identification, name, address, telephone number, and email.
- The system must record staff details, including their staff ID, name, phone number, and email.
- A customer can buy many artworks, and an artwork can be bought by only one customers. Each sale transaction must also record the date, time, and total sale amount.
- Each staff member manages at least one sales transaction, and a transaction can only be managed by one staff member.
- A manager, appointed from among the staff, must monitor at least one staff member. Each staff member can only be monitored by one manager.

b) Convert the ERD you have constructed in (a) into Chen Notation. (7.5 marks)